Practicum No.7 **Date: 20/2/2026**

Visit Report on a CSE/AIML/DSCS/DS Incubator or Support Agency

I. Practical Significance

The purpose of this visit was to understand the functioning, services, and practical relevance of an Entrepreneurship Support Agency / CSE–AIML–DSCS–DS Incubator for nurturing early-stage startups. As part of the practicum requirement, the visit aimed to expose students to real-world innovation ecosystems, startup support mechanisms, funding pathways, and technology incubation processes.

The selected agency/incubator acts as a catalyst for converting ideas into viable products through structured mentoring, access to infrastructure, and business development support. The visit provided practical insights into how startups integrate emerging technologies such as Artificial Intelligence (AI), Machine Learning (ML), Data Science (DS), and Cloud Systems (CS).

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- ***Develop*** a pitch deck, project proposals for small/medium scale startups logically and scientifically.

a) ***Interpret*** competitive dynamics shaping CSE-focused high-tech and software-driven markets.
b) ***Evaluate*** commercial viability of CSE products such as AI models, software applications, analytics platforms, and cloud-native solutions.
c) ***Innovate*** strategic direction for new ventures in advanced CSE domains including generative AI, distributed systems, and intelligent automation.
d) ***Practice*** professional ethics in data collection, system design, and user interaction within digital ecosystems.

III. Relevant CO(s)

- **CO3:** ***Formulate*** a startup proposal for the chosen idea through entrepreneurial support systems.

IV. Practical Outcome (PrO)

- ***Present*** a report from a visit to an entrepreneurship support agency or CSE, AIML, DSCS & DS incubator for the ***identified*** start-up.

V. Related ADO(s) (keep as such)

a) ***Practice*** professional ethics.
b) ***Function*** effectively as a team member/leader.
c) ***Practice*** integrity and honesty.

VI. Minimum Underpinning Theory

The practicum visit is grounded in essential theoretical concepts from computing, innovation management, and entrepreneurial development. Fundamental theories in software engineering, cloud-native architectures, AI/ML model pipelines, and data-centric system design provide the technical base for understanding how digital products are developed and scaled within an incubation environment. Innovation theories such as the Technology Readiness Levels (TRLs), Lean Startup methodology, Design Thinking, and Agile Development frameworks explain how early-stage ventures refine ideas, validate market needs, and build minimum viable products. Market dynamics theories, including competitive analysis, customer segmentation, and value proposition modelling, support the evaluation of startup feasibility and positioning. Additionally, principles of professional ethics, responsible AI, data privacy, and secure system design underpin the ethical guidelines observed in digital ecosystems. Together, these theories provide the foundational knowledge required to interpret incubation processes and analyze the strategies adopted by CSE, AIML, DSCS, and DS-based startups during the visit.

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and documentation	1 per team
2	Internet access	Trend research	As required
3	Brainstorming sheets/chart paper	For idea listing	1 set
4	Reference materials	Magazines, journals, websites	1 per team

IX. Safety Precautions

a) ***Ensure*** ideas are original and not plagiarized**.**
b) ***Avoid*** using confidential or proprietary data.
c) ***Maintain*** ethical considerations in proposed ventures.
d) ***Respect*** social, cultural, and privacy boundaries.

X. Procedure

1. ***Form*** a team and select the domain of interest (CSE/AIML/DSCS/DS).
2. ***Observe*** current problems or opportunities in the domain.
3. ***Conduct*** brainstorming / SCAMPER / mind-mapping sessions.
4. ***List*** all raw ideas generated.
5. ***Filter*** out impractical, unethical, or repetitive ideas.
6. ***Refine*** shortlisted ideas
   a. ***Target users***
   b. ***Market problem***
   c. ***Technology required***
   d. ***Value proposition***
7. ***Document*** the ideas in a report or prepare a slide presentation.
8. ***Submit*** to faculty for evaluation.

XI. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1	Laptop	HP	Used for research and documentation	1	-
2	Internet Access	Wi-Fi	Used for collecting information about incubator	1	-
3	Notepad	Classmate	Used for writing observations and notes	1	-
4	Reference Materials	Journals	Used for gathering startup and incubator information	1	-

XII. Actual Procedure Followed

1. First, we visited the Karunya Innovation and Incubation Centre (KIIC) available on campus to understand how startups are supported.
2. During the visit, we observed the facilities provided by the incubator, such as workspace, technical labs, mentoring programs, and startup support infrastructure.
3. We collected information about how startup ideas are developed and converted into real products through incubation programs.
4. We interacted with incubator staff and learned about the mentoring, training, and funding opportunities available for student startups.
5. We also discussed how emerging technologies such as Artificial Intelligence, Data Science, and Cloud Computing are used in startup solutions.
6. The information collected was noted and analyzed to understand how our idea Digital Attendance and Productivity Tracker could benefit from incubation support.
7. Finally, the observations and insights gained from the visit were documented in the practicum report.

XIII. Observations and Data

S. No.	Support Agency identified	Data/ Remarks
1	Incubator Name	Karunya Innovation and Incubation Centre (KIIC)
2	Startup Support	Provides mentoring, training, and business guidance
3	Infrastructure	Offers workspace, labs, and technical resources
4	Technology Focus	AI, ML, Cloud Computing, Data Science
5	Relevance to Project	Can support development of the Digital Attendance and Productivity Tracker

XIV. Results/Observations

1. The incubator provides an environment where students and innovators can convert ideas into real startup products.
2. Facilities such as mentoring, infrastructure, and technical support help entrepreneurs develop and test their solutions.
3. The incubation ecosystem encourages the use of modern technologies like AI, cloud computing, and data analytics to build innovative products.

XV. Interpretation of Results

1. The visit helped us understand how incubators support **early-stage startups by providing guidance, technical resources, and networking opportunities**.
2. For our idea, **Digital Attendance and Productivity Tracker**, incubation support could help in developing the prototype, testing the system, and improving the product before launching it.
3. This shows that incubation programs play an important role in helping students **transform innovative ideas into practical solutions**.

XVI. Conclusions

1. The visit to the incubator provided valuable insights into the startup ecosystem and the support systems available for new entrepreneurs.
2. It highlighted the importance of mentorship, technical infrastructure, and funding in building successful technology startups.

3. The knowledge gained from this visit will help in improving our idea Digital Attendance and Productivity Tracker and understanding how incubation programs can support future entrepreneurial ventures.

XVII. Suggested Practicum Related Questions

1. **Evaluate** how internal technical capabilities (such as programming expertise, algorithmic skills, data engineering capacity, and cloud deployment abilities) influence the strategic position of a CSE/AIML/DSCS/DS in the market.
2. **Evaluate** the role of emerging market trends—such as generative AI, privacy-preserving analytics, automation, cybersecurity modernization, and edge computing—in shaping external opportunities for computing-based startups.
3. **Assess** how operational limitations (compute resources, model training time, data availability, infrastructure cost, or cybersecurity risks) and data-related constraints (quality, volume, privacy, and compliance requirements) affect the viability of the products.
4. **Examine** competitive threats faced by AI-based applications, cybersecurity platforms, data-driven services, or cloud-native solutions—considering factors such as large tech competitors, open-source tools, rapid technological change, regulatory risks, and customer adoption barriers.
5. **Interpret** the insights from the incubator visit can guide MVP (Minimum Viable Product) development, technology stack selection, customer validation, and long-term venture strategy for computing startups.

XVIII References / Suggestions for further reading (keep as such)

a) Venture Meets Mission: Aligning People, Purpose, and Profit to Innovate and Transform Society, Arun Gupta, Gerard George, Thomas J. Fewer, Stanford University Press, Stanford, California, 2024, ISBN: 978-1503636283
b) So You Want to Start a Business: Eight Steps to Startup Success, John B. Vinturella, Business Expert Press, New York, USA, 2024, ISBN: 978-1637425855
c) Business Planning, Franklin A. Gevurtz, Foundation Press (West Academic), St. Paul, Minnesota, 2024, ISBN: 978-1685612764

XIX Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the ***process-related* skills** and ***product-related* skills** will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Identification of domain & problems	3	
2	Use of structured idea generation techniques	4	
3	Team collaboration & creativity	4	
4	Feasibility evaluation	3	
5	Ethical originality & clarity	4	
Product-related Skills: 12 Marks - 40%			
1	Quality & novelty of ideas	2	
2	Logical presentation & organization	2	
3	Clarity of findings & conclusions	3	
4	Report/presentation submission	3	
5	Innovation potential & market relevance	2	
	Total	**30**	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	**Neavis Rishva. J**	**URK25CS1236**

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Signature with Date
Process-Related skills (18)	Product-Related skills (12)	Total (30)		